

Control and Optimization for Modern Energy Systems

A battery plant tour and an afternoon of technical talks at the University of Hawai'i at Mānoa

Kapolei Energy Storage
8:00 am site tour

UH Mānoa
1:00 pm talks

DATE	TIME	VENUE	REGISTRATION
Monday, December 14, 2026 Adjacent to IEEE CDC 2026, Dec. 15–18	8:00 am – 6:00 pm Tour, then talks	UH Mānoa campus Honolulu, O'ahu	No fee Separate event from CDC

About the workshop

The IEEE CSS Technical Committee on Energy Systems (TC-ES) is organizing a full-day technical workshop that brings together researchers and industry practitioners who work on control and optimization for modern energy systems.

A university campus in Hawai'i is a fitting setting for this workshop, because the islands operate some of the most renewable-rich, inverter-dominated grids in the world. Many of the problems that engineers in Hawai'i are solving today inform global solutions for grid-forming control, distributed flexibility, large new loads, and reliability under deep uncertainty.

Program

- Grid-forming and inverter-based resource control
- Next-generation grid codes and ancillary services
- Grid-aware demand response and DER coordination
- Congestion and voltage management
- Virtual power plants and market participation
- Data centers and AI compute as flexible grid assets
- Operational perspectives from Hawaiian Electric

Confirmed speakers



Ken Aramaki
Director, Integrated
Grid Planning,
Hawaiian Electric



Mark Glick
Chief Energy Officer,
State of Hawai'i



Daisy Green
University of Hawai'i
at Mānoa



Bai Cui
Iowa State University



Sairaj Dhople
University of Minnesota



Vladimir Dvorkin
University of Michigan,
Ann Arbor



Verena Häberle
ETH Zürich



**Patricia
Hidalgo-Gonzalez**
UC San Diego



Sorin Olaru
CentraleSupélec



Siva Seetharaman
Purdue University



Johanna Vorwerk
Technical University of
Denmark (DTU)

WORKSHOP ORGANIZERS

Daisy Green, Johanna Mathieu, Mads Almassalkhi (Chair, IEEE CSS TC-ES), and Patrick Panciatici

Questions: dgreen2@hawaii.edu and jlmath@umich.edu

Schedule



8:00 AM – 1:00 PM

Tour of the Kapolei Energy Storage facility

Buses take participants from UH Mānoa to Kapolei on western O'ahu to visit a 185 MW / 565 MWh grid-forming battery plant.



1:00 PM – 6:00 PM

Technical talks at UH Mānoa

Invited talks from universities, Hawaiian Electric, and the State of Hawai'i.

Logistics

- UH Mānoa is a 15-minute drive or a 50-minute walk from the CDC hotel, the Hilton Hawaiian Village.
- The workshop is separate from CDC and has no registration fee.



Program and updates
[madsalma.github.io/
hawaii2026.html](https://madsalma.github.io/hawaii2026.html)